

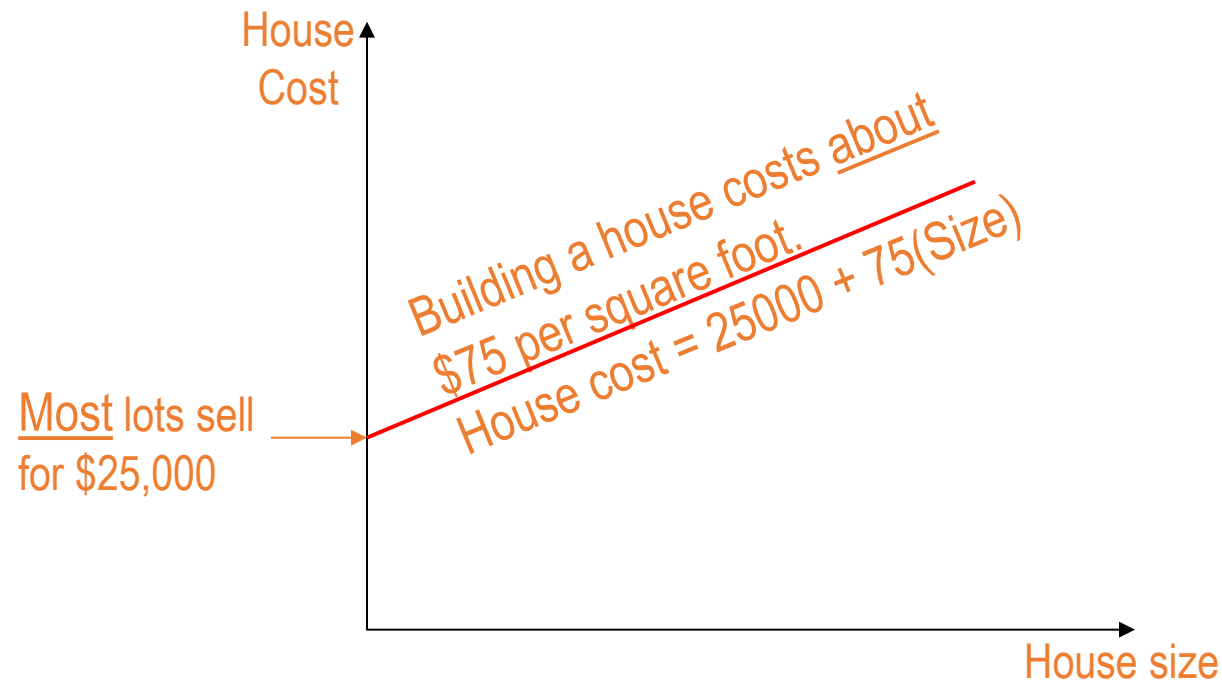
Linear Regression

Introduction

- The motivation for using the technique:
 - Forecast the value of a dependent variable (Y) from the value of independent variables ($X_1, X_2, \dots, X_k$).
 - Analyze the specific relationships between the independent variables and the dependent variable.

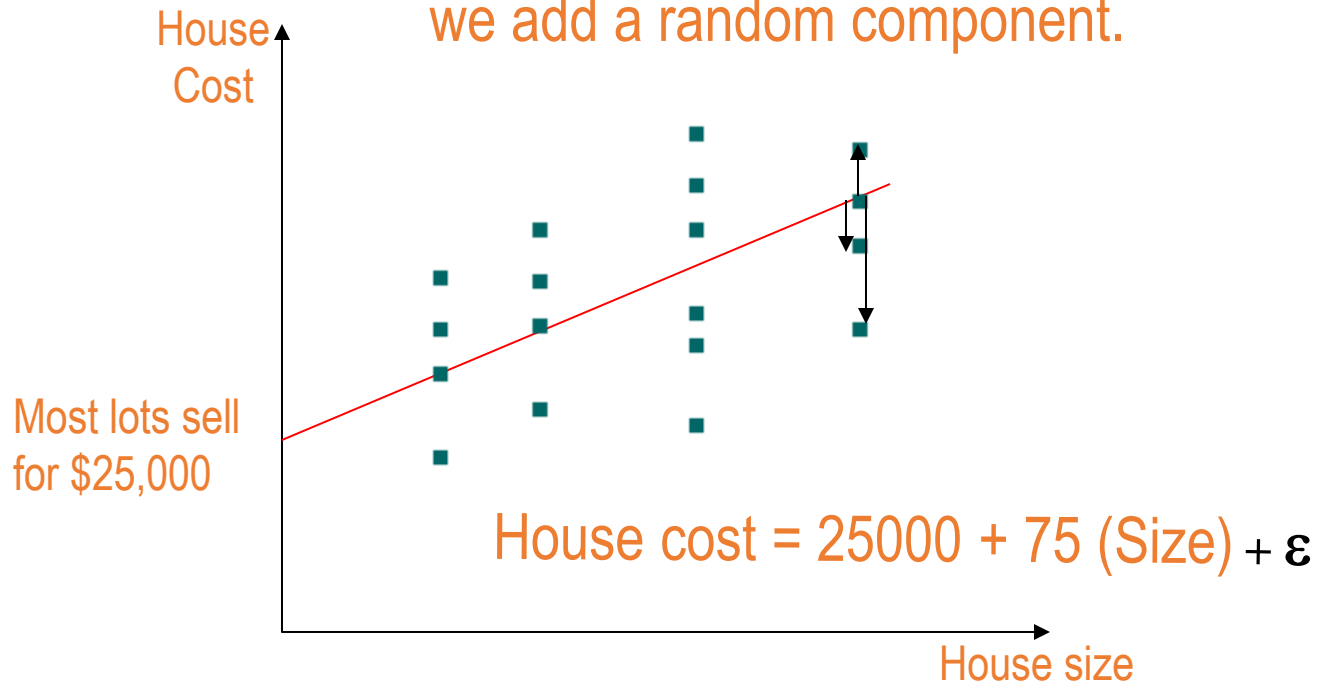
The Model

The model has a deterministic and a probabilistic components



However, house cost vary even among same size houses!

Since cost behave unpredictably,
we add a random component.



- The first order linear model

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

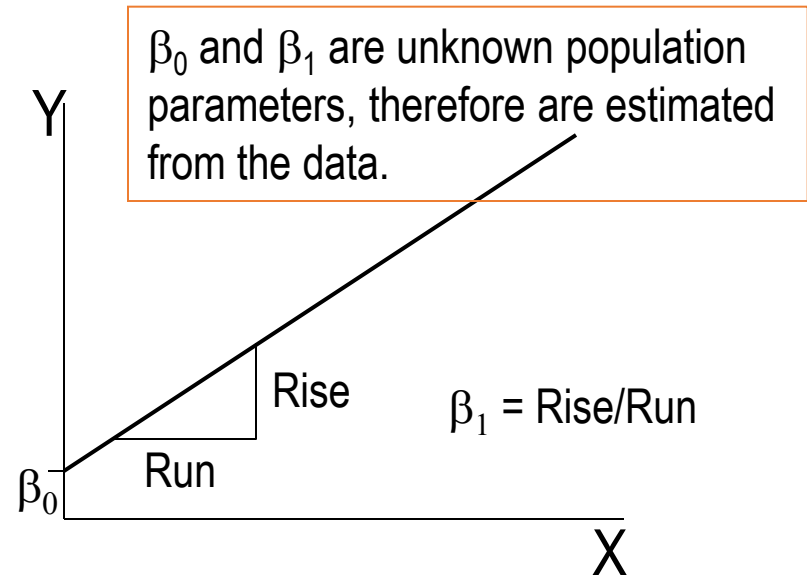
Y = dependent variable

X = independent variable

β_0 = Y-intercept

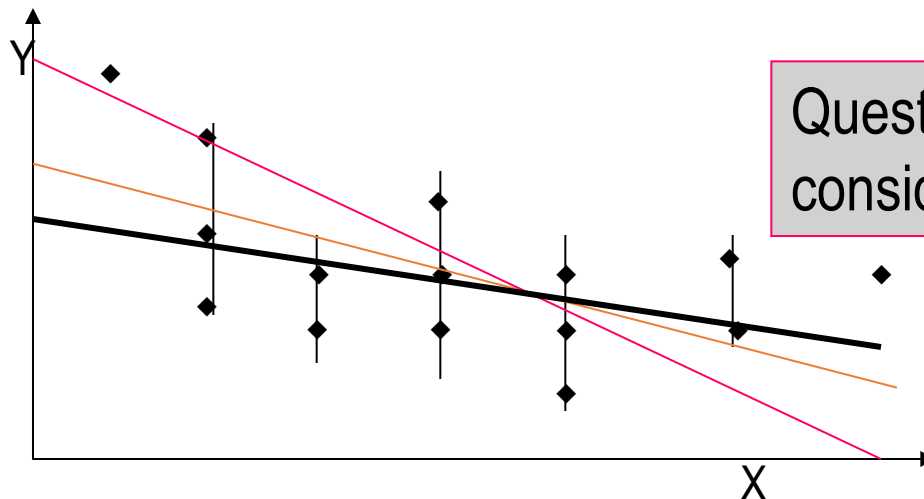
β_1 = slope of the line

ε = error variable



Estimating the Coefficients

- The estimates are determined by
 - drawing a sample from the population of interest,
 - calculating sample statistics.
 - producing a straight line that cuts into the data.



Question: What should be considered a good line?

The Least Squares (Regression) Line

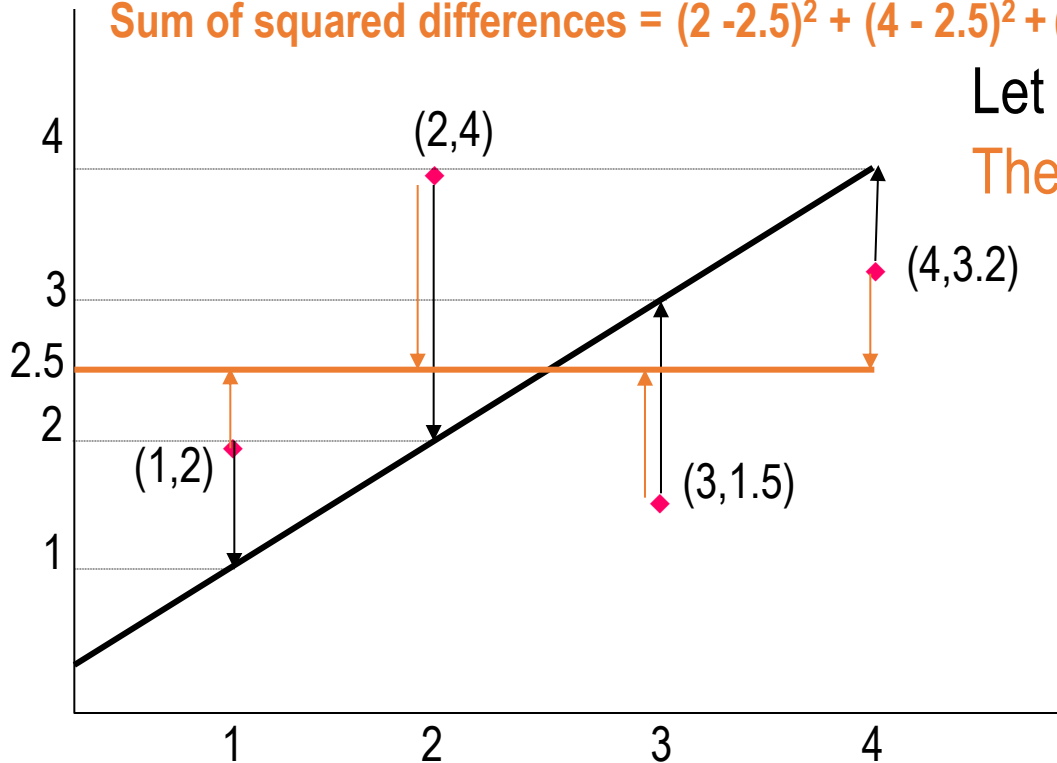
A good line is one that minimizes the sum of squared differences between the points and the line.

Sum of squared differences = $(2 - 1)^2 + (4 - 2)^2 + (1.5 - 3)^2 + (3.2 - 4)^2 = 6.89$

Sum of squared differences = $(2 - 2.5)^2 + (4 - 2.5)^2 + (1.5 - 2.5)^2 + (3.2 - 2.5)^2 = 3.99$

Let us compare two lines

The second line is horizontal



The smaller the sum of squared differences the better the fit of the line to the data.

The Estimated Coefficients

To calculate the estimates of the line coefficients, that minimize the differences between the data points and the line, use the formulas:

$$b_1 = \frac{\text{cov}(X, Y)}{s_X^2} \left(= \frac{s_{XY}}{s_X^2} \right)$$

$$b_0 = \bar{Y} - b_1 \bar{X}$$

The regression equation that estimates the equation of the first order linear model is:

$$\hat{Y} = b_0 + b_1 X$$

The Simple Linear Regression Line

- Example 1
 - A car dealer wants to find the relationship between the odometer reading and the selling price of used cars.
 - A random sample of 100 cars is selected, and the data recorded.
 - Find the regression line.

Car	Odometer	Price
1	37388	14636
2	44758	14122
3	45833	14016
4	30862	15590
5	31705	15568
6	34010	14718
.	Independent	Dependent
.	variable X	variable Y
.	.	.

- Solution

- Solving by hand: Calculate a number of statistics

$$\bar{X} = 36,009.45; \quad s_X^2 = \frac{\sum (X_i - \bar{X})^2}{n-1} = 43,528,690$$

$$\bar{Y} = 14,822.823; \quad \text{cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n-1} = -2,712,511$$

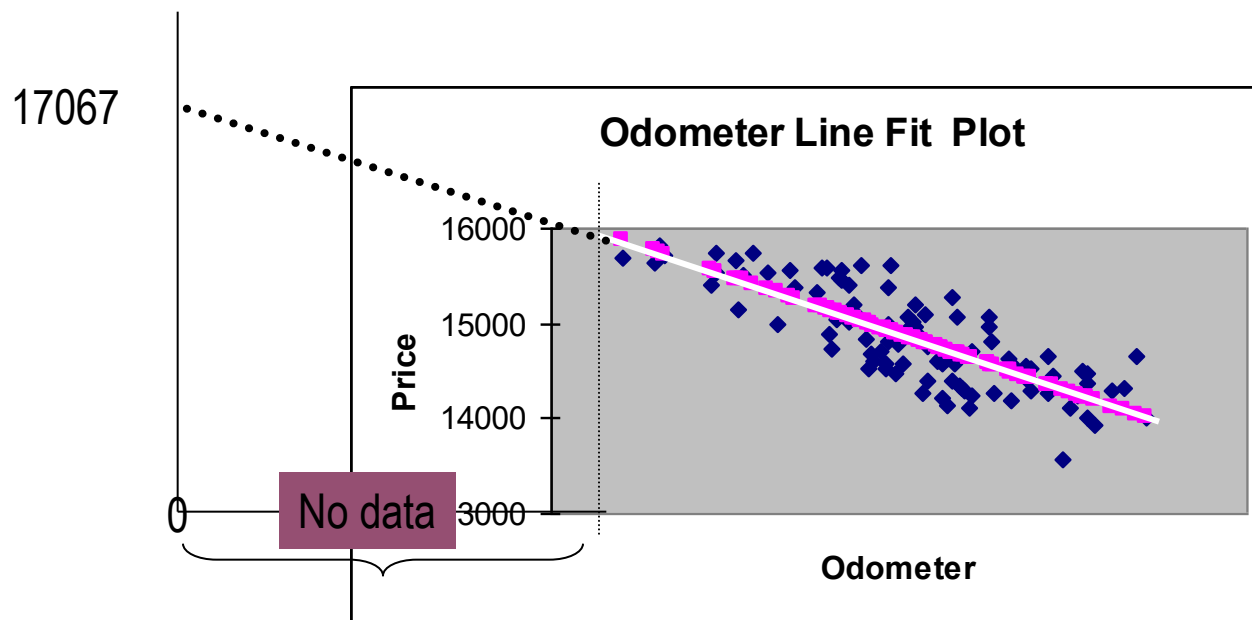
where $n = 100$.

$$b_1 = \frac{\text{cov}(X, Y)}{s_X^2} = \frac{-1,712,511}{43,528,690} = -.06232$$

$$b_0 = \bar{Y} - b_1 \bar{X} = 14,822.82 - (-.06232)(36,009.45) = 17,067$$

$$\hat{Y} = b_0 + b_1 X = 17,067 - .0623X$$

Interpreting the Linear Regression -Equation



$$\hat{Y} = 17,067 - .0623X$$

The intercept is $b_0 = \$17067$.

Do not interpret the intercept as the
"Price of cars that have not been driven"

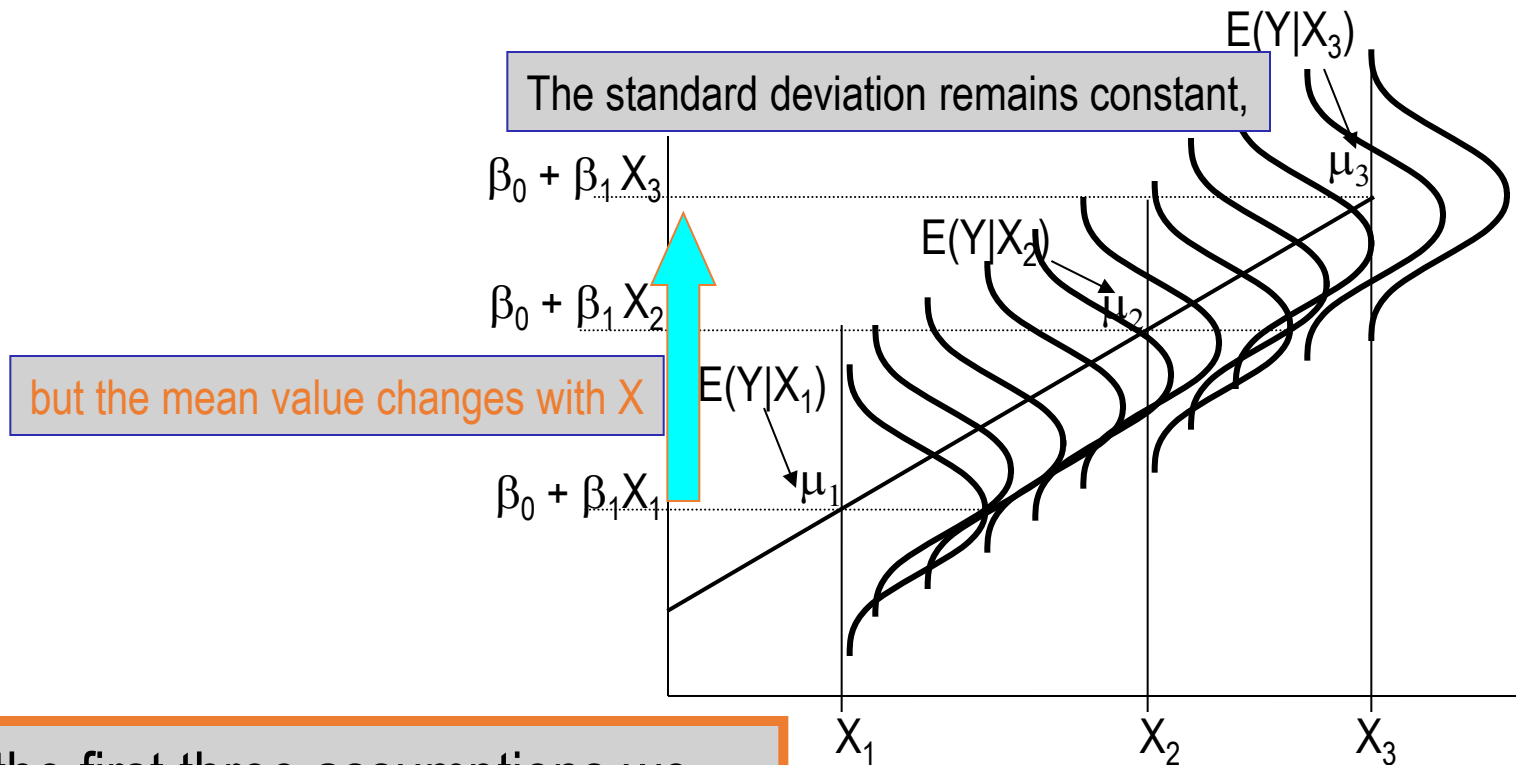
This is the slope of the line.

For each additional mile on the odometer,
the price decreases by an average of \$0.0623

Error Variable: Required Conditions

- The error ε is a critical part of the regression model.
- Four requirements involving the distribution of ε must be satisfied.
 - The probability distribution of ε is normal.
 - The mean of ε is zero: $E(\varepsilon) = 0$.
 - The standard deviation of ε is σ_ε for all values of X .
 - The set of errors associated with different values of Y are all independent.

The Normality of ε



From the first three assumptions we have: Y is normally distributed with mean $E(Y) = \beta_0 + \beta_1 X$, and a constant standard deviation σ_ε

Assessing the Model

- The least squares method will produce a regression line whether or not there is a linear relationship between X and Y .
- Consequently, it is important to assess how well the linear model fits the data.
- Several methods are used to assess the model. All are based on the sum of squares for errors, SSE.

Sum of Squares for Errors

- This is the sum of differences between the points and the regression line.
- It can serve as a measure of how well the line fits the data. SSE is defined by

$$\text{SSE} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

– A shortcut formula

$$\text{SSE} = (n - 1)s_Y^2 - \frac{[\text{cov}(X, Y)]^2}{s_X^2}$$

Standard Error of Estimate

The mean error is equal to zero.

If se is small the errors tend to be close to zero (close to the mean error). Then, the model fits the data well.

Therefore, we can, use se as a measure of the suitability of using a linear model.

An estimator of se is given by se

Standard Error of Estimate

$$s_e = \sqrt{\frac{SSE}{n-2}}$$

- Example 2

- Calculate the standard error of estimate for Example 1, and describe what does it tell you about the model fit?

- Solution

$$s_Y^2 = \frac{\sum (Y_i - \hat{Y}_i)^2}{n-1} = 259,996$$

$$SSE = (n-1)s_Y^2 - \frac{[\text{cov}(X, Y)]^2}{s_X^2} = 99(259,996) - \frac{(-2,712,511)^2}{43,528,690} = 9,005,450$$

Calculated before

$$s_\varepsilon = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{9,005,450}{98}} = 303.13$$

It is hard to assess the model based on s_ε even when compared with the mean value of Y.

$$s_\varepsilon = 303.1 \quad \bar{y} = 14,823$$